# Arcs

SEE ALSO
❮ 56–59 Ratio and proportion
❮ 138–139 Circles
❮ 140–141 Circumference and diameter

AN ARC IS A SECTION OF A CIRCLE'S CIRCUMFERENCE. ITS LENGTH CAN BE FOUND USING ITS RELATED ANGLE AT THE CENTER OF THE CIRCLE.

## What is an arc?

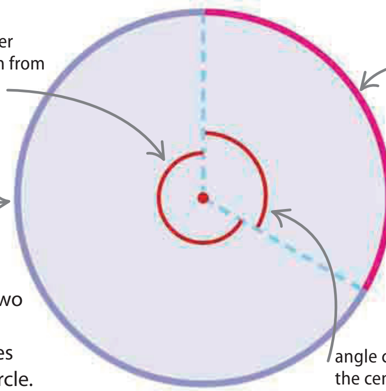
An arc is a part of the circumference of a circle. The length of an arc is in proportion with the size of the angle made at the center of the circle when lines are drawn from each end of the arc. If the length of an arc is unknown, it can be found using the circumference and this angle. When a circle is split into two arcs, the bigger is called the "major" arc, and the smaller the "minor" arc.

formula for finding the length of an arc

$$\frac{\text{arc length}}{\text{circumference}} = \frac{\text{angle at center}}{360^\circ}$$

total length of circle's edge

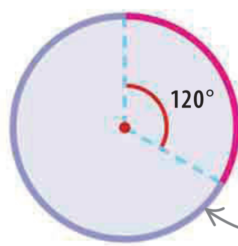
360° in a circle




▷ **Arcs and angles**
This diagram shows two arcs: one major, one minor, and their angles at the center of the circle.

## Finding the length of an arc

The length of an arc is a proportion of the whole circumference of the circle. The exact proportion is the ratio between the angle formed from each end of the arc at the center of the circle, and 360º, which is the total number of degrees around the central point. This ratio is part of the formula for the length of an arc.




◁ **Find the arc length**
This circle has a circumference of 10 cm. Find the length of the arc that forms an angle of 120º at the center of the circle.

**Take the formula** for finding the length of an arc. The formula uses the ratios between arc length and circumference, and between the angle at the center of the circle and 360° (total number of degrees).

$$\frac{\text{arc length}}{\text{circumference}} = \frac{\text{angle at center}}{360^\circ}$$

**Substitute the numbers that are known** into the formula. In this example, the circumference is known to be 10 cm, and the angle at the center of the circle is 120º; 360º stays as it is.

$$\frac{\text{arc length}}{10} = \frac{120}{360}$$

this side has been multiplied by 10 to leave arc length on its own (÷10 × 10 cancels out)

this side has also been multiplied by 10 because what is done to one side must be done to the other

**Rearrange the equation** to isolate the unknown value—the arc length—on one side of the equals sign. In this example the arc length is isolated by multiplying both sides by 10.

$$\text{arc length} = \frac{10 \times 120}{360}$$

3.333... is rounded to 2 decimal places

**Multiply** 10 by 120 and divide the answer by 360 to get the value of the arc length. Then round the answer to a suitable number of decimal places.

$$C = 3.33\text{cm}$$